DRIFTLESS — Database Schema & Data Flow Architecture

1. Database Overview & Migration History

DRIFTLESS utilizes **SQLite 3** as its primary persistence engine (`db.sqlite3`), supported natively by Django's database abstraction layer.

Applied Migrations (`dashboard/core/migrations/`):

- `0001_initial` : Created initial base models (`UsageLog` , `UserSettings`).
- `0002_usagelog_category_usersettings` : Added `category` column to `UsageLog` and expanded `UserSettings` .
- `0003_userprofile` : Created `UserProfile` model extending Django's auth `User` with phone and persona.
- `0004_alter_userprofile_persona` : Updated persona choices to support `student` , `professional` , `influencer` , `creator` , and `other` .
- `0005_usersettings_allowed_sites_and_more` : Added `SiteLimit` , `CategoryOverride` , `FocusSession` , unique constraints, and indexes.

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2. Entity-Relationship (ER) Schema

```
erDiagram
    auth_user ||──|| core_userprofile : "extends (1:1)"
    auth_user ||──|| core_usersettings : "configures (1:1)"
    auth_user ||──o{ core_usagelog : "accumulates (1:N)"
    auth_user ||──o{ core_sitelimit : "enforces (1:N)"
    auth_user ||──o{ core_categoryoverride : "overrides (1:N)"
    auth_user ||──o{ core_focussession : "completes (1:N)"
    auth_user ||──o| authtoken_token : "authenticates (1:1)"

    auth_user {
        int id PK
        varchar_150 username "UNIQUE"
        varchar_254 email
        varchar_128 password
        boolean is_active
        datetime date_joined
    }

    core_userprofile {
        int id PK
        int user_id FK "UNIQUE"
        varchar_32 phone
        varchar_20 persona "student|professional|influencer|creator|other"
    }

    core_usersettings {
        int id PK
        int user_id FK "UNIQUE"
        boolean focus_mode_active
        integer daily_limit_minutes "DEFAULT 60"
        integer focus_goal_minutes "DEFAULT 180"
        integer streak_days "DEFAULT 0"
        boolean require_challenge "DEFAULT FALSE"
        text blocked_sites
        text allowed_sites
        datetime updated_at
    }

    core_usagelog {
        int id PK
        int user_id FK
        varchar_255 website_url
        integer time_spent_seconds "DEFAULT 0"
        varchar_20 category "Productivity|Social|Entertainment|Other"
        date date
        datetime created_at
        datetime updated_at
    }

    core_sitelimit {
        int id PK
        int user_id FK
```

```
        varchar_255 domain
        integer daily_limit_minutes "DEFAULT 30"
        boolean enabled "DEFAULT TRUE"
        datetime created_at
        datetime updated_at
    }

    core_categoryoverride {
        int id PK
        int user_id FK
        varchar_255 domain
        varchar_30 category
        datetime created_at
        datetime updated_at
    }

    core_focussession {
        int id PK
        int user_id FK
        integer duration_minutes "DEFAULT 25"
        datetime started_at
        boolean completed "DEFAULT TRUE"
    }

    authtoken_token {
        varchar_40 key PK
        int user_id FK "UNIQUE"
        datetime created
    }
```

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3. Detailed Model Field Specifications

3.1. `core_usagelog`

- **Primary Key:** `id` (`BigAutoField`, Auto-increment).
- **Foreign Key:** `user_id` referencing `auth_user(id)` with `on_delete=CASCADE`.
- **Fields:**
- `website_url` (`CharField(255)`): Sanitized domain name (e.g. `github.com`).
- `time_spent_seconds` (`PositiveIntegerField`): Accumulated dwell time for that calendar date.
- `category` (`CharField(20)`): Classification label (`Productivity` , `Social` , `Entertainment` , `Other`).
- `date` (`DateField`): The specific calendar date of the dwell activity.
- **Constraints & Indexes:**
- `unique_together = ('user', 'website_url', 'date')` guarantees that each website has at most one row per user per calendar day.

- `Index(fields=['user', 'date'])` accelerates 7-day range scans for weekly summaries.

3.2. `core_sitelimit`

- **Unique Constraint:** `unique_together = ('user', 'domain')`.
- **Fields:**
- `domain` (`CharField(255)`): Target sanitized hostname.
- `daily_limit_minutes` (`PositiveIntegerField`, default `30`): Daily time cap in minutes.
- `enabled` (`BooleanField`, default `True`): Toggles enforcement of the limit.

3.3. `core_categoryoverride`

- **Unique Constraint:** `unique_together = ('user', 'domain')`.
- **Fields:**
- `domain` (`CharField(255)`): Overridden hostname.
- `category` (`CharField(30)`): Target category choice.

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4. End-to-End Data Flow (Browser Storage ↔ Backend Database)

```
sequenceDiagram
    autonumber
    participant ActiveTab as User Tab (e.g. youtube.com)
    participant SW as Service Worker (background.js)
    participant Storage as chrome.storage.local
    participant API as Django Ingestion (/api/usage/)
    participant DB as SQLite (UsageLog)
    participant SummaryAPI as Django Analytics (/api/summary/)
    participant Dashboard as Web Dashboard (dashboard.html)

    ActiveTab->>SW: Navigation & Active Dwell
    SW->>Storage: Increment siteTimes['youtube.com'] += 60s
    SW->>Storage: Increment unsyncedDeltas['youtube.com'] += 60s

    Note over SW,API: 60s Periodic Sync Alarm Triggered
    SW->>Storage: Read unsyncedDeltas
    SW->>API: POST /api/usage/ [{ website_url: "youtube.com", time_spent_seconds: 60, date:
"2026-10-26" }]
    API->>DB: UsageLog.objects.get_or_create(user, "youtube.com", date)
    API->>DB: time_spent_seconds += 60, save()
    API-->>SW: HTTP 200 OK { created: 0, updated: 1 }
    SW->>Storage: unsyncedDeltas['youtube.com'] = 0

    Note over Dashboard,SummaryAPI: Dashboard Page Loaded or Auto-Refreshed
    Dashboard->>SummaryAPI: GET /api/summary/
    SummaryAPI->>DB: Query 7-day logs (UsageLog.objects.filter(date__gte=week_ago))
    SummaryAPI->>DB: Calculate strict streak & focus efficiency
    SummaryAPI-->>Dashboard: Return { weekly: [...], daily: [...], stats: {...} }
    Dashboard->>Dashboard: Chart.js updates canvas; KPI metrics rendered
```

5. Storage Partitioning: Browser Local vs. Backend Database

Metric / Attribute	Browser-Local Storage (`chrome.storage.local`)	Backend Database (`db.sqlite3`)
Storage Lifetime	Resets daily at midnight; cleared if extension is uninstalled.	Permanent historical record across months and years.
Tracking Granularity	Sub-minute raw second increments (`siteTimes`).	Daily aggregated seconds (`UsageLog.time_spent_seconds`).
Sync Buffer	Unsent delta queue (`unsyncedDeltas`) buffered during offline intervals.	Clean committed state.
Enforcement States	3-tier escalation flags (`escalationState[domain].nudged/alerted/blocked`).	`SiteLimit` rules and threshold configurations.

Metric / Attribute	Browser-Local Storage (`chrome.storage.local`)	Backend Database (`db.sqlite3`)
Focus Mode State	Dynamic DNR rule status (`focusModeActive` , `focusTimerEndTime`).	Persistent user preference flag (`UserSettings.focus_mode_active`).
Authentication	Bearer API token string (`apiToken`).	`authtoken_token` table linking token key to `auth_user` .